Xudong Wu

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Interested in Reinforcement Learning for embodied AI/LLM.

Education

The University of Hong Kong (HKU) — **Hong Kong SAR**
Doctor of Philosophy, — *09/2025 – 08/2029 (Expected)*

Supervisor: Prof. Jiayu Chen (Primary), Prof. Vaneet Aggarwal (Co-supervisor), Prof. Wenjie Huang (Co-supervisor).
Tentative Topic: Reinforcement Learning, embodied AI, Large language model.

University of Edinburgh (UoE) — **Edinburgh, UK**
Bachelor of Science (Honours) in Mathematics and Statistics, — *09/2023 – 07/2025*

Average Score: 78/100 (*First Class Honours*).
Note: In the UK grading system, a score of 70+ is equivalent to GPA 4.0.

University of California, Irvine (UCI) — **Irvine, USA**
Research Assistant, Donald Bren School of ICS, — *06/2024 – 09/2024*

Supervisor: Prof. Chen Li (IEEE Fellow).
Project: LLM-based Agents for Automated Workflow Analysis on Texera Platform.

Dalian University of Technology (DUT, 985/211) — **Dalian, China**
Bachelor of Science in Information and Computing Science, — *09/2021 – 06/2023*

Overall GPA: 3.99 (Top 5%).

Research Experience

A Comparative Study of Simulation-Based Inference Algorithms — **Edinburgh, UK**
Honours Dissertation/Final Year Project, Advisor: Dr. Amanda Lenzi — *08/2024–05/2025*

- Conducted the first systematic comparison of BayesFlow, Sequential Neural Likelihood (SNL), and Affine Flow Matching (AFM) in structured high-dimensional inference tasks, particularly under spatial disease mapping scenarios.
- Designed benchmark experiments on Gaussian posterior recovery and extended them to Poisson–CAR models with strong spatial correlation, leveraging variational inference and flow-based methods.
- Discovered a trade-off in joint parameter inference accuracy and evaluated posterior calibration using rank-based ECDF and recovery line diagnostics.
- Identified AFM's robustness in modeling global precision parameters across varying spatial smoothness levels and discussed future directions with GNN/Transformer-based architectures.

Optimizing Texera, a Large-Scale Workflow System for LLM-Based Data Analysis — **Irvine, USA**
Summer Research Project, Advisor: Prof. Chen Li (IEEE Fellow) — *06/2024–08/2024*

- Contributed to the development of Texera, a large-scale workflow engine for data analysis, focusing on backend architecture, HTML report generation, and real-time visualization components.
- Designed and implemented a modular LLM-based agent ("Storyteller AI") to autonomously interpret pipeline outputs, generate structured summaries, and provide GPT-4 powered natural language reports.
- Integrated multi-agent prompting strategies and reflective LLM chains to automate the end-to-end workflow reporting process.
- Improved system scalability and inference throughput through code-level optimization and multi-threaded job execution, enhancing the platform's responsiveness under heavy user load.

The Economic Impact of Uncertainty During the COVID-19 Pandemic — **Edinburgh, UK**
Research Assistant — *02/2024–05/2024*

- Implemented core probabilistic algorithms such as Monte Carlo sampling to model economic uncertainty, exploring inference techniques under limited and noisy observations.
- Studied the application of Bayesian methods in real-world data settings, focusing on decision-making under uncertainty—laying the foundation for later work in simulation-based inference and sequential learning.
- Developed reproducible statistical pipelines with R and Python to support transparent modeling, calibration diagnostics, and visual communication of inference results.

Project Experience

Self-Rewarding Framework for Medical LLMs **Edinburgh, UK**
Project Leader in Group Project, Machine Learning Practical *01/2025–03/2025*

- Designed a novel self-rewarding framework for aligning medical LLMs without human annotations, using Direct Preference Optimization (DPO) and dynamic LLM-as-a-Judge prompts generated by ChatGPT-4o.
- Implemented a hierarchical evaluation pipeline including iterative self-question generation, multi-response generation, self-judgment scoring, and adaptive judge refinement.
- Conducted experiments on PubMedQA and MedMCQA, demonstrating significant improvements in structure, specificity, empathy, and completeness of model responses compared to supervised fine-tuning.
- Identified and analyzed failure modes in high-difficulty medical tasks due to reward misspecification and hallucination accumulation, providing insights into the task-dependent limitations of self-reinforcing loops.

Advanced Analysis and Experimental Optimization of Neural Network Architectures **Edinburgh, UK**
Individual project in Machine Learning Practical *09/2024-12/2024*

- Designed and executed a series of controlled experiments to investigate the vanishing gradient problem, revealing its impact on training dynamics and proposing effective mitigation strategies.
- Conducted extensive performance benchmarking across various configurations, demonstrating a measurable increase in accuracy and robustness, particularly in models incorporating residual connections.
- Proposed novel experimental setups to further explore the behavior of Batch Normalization and Residual Connections, providing insights into their synergistic effects on training stability and feature representation.
- Synthesized experimental results to derive actionable recommendations for optimizing neural network architectures.

Advanced Statistical Modeling and Bayesian Inference **Edinburgh, UK**
Individual Project in Course: "Statistical Computing" *02/2024–05/2024*

- Developed Bayesian linear models for accurate 3D material usage estimation, leveraging both classical and probabilistic modeling techniques to improve predictive stability under uncertainty.
- Applied Bayesian inference with prior integration and Monte Carlo refinement to iteratively update model beliefs—laying foundational skills for posterior shaping in simulation-based learning.
- Conducted rigorous evaluation using k-fold cross-validation and uncertainty-aware metrics, reinforcing understanding of model reliability for adaptive decision systems.

Honors and Awards

- Outstanding Student of DUT, 09/2022-06/2023.
- First-Class Scholarship of DUT (Top 5%), 09/2022-06/2023.
- International Study Scholarship of DUT, 09/2022-06/2023.
- Outstanding Student of DUT, 09/2021-06/2022.
- Second-Class Scholarship of DUT (Top 5%-20%), 09/2021-06/2022.

Skills

Programming: Python (proficient), R, C/C++, SQL, Scala, MATLAB
Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn
Tools & Platforms: Git, Docker, OpenAI API, LaTeX, Linux (Bash), Microsoft Office Suite
Languages: English (Fluent), Mandarin (Native), Cantonese (Beginner)